

NanoSafe Analyzer — Research & Cytotoxicity Report

In Vitro Evaluation of ZnO Nanoparticle Biocompatibility | ISO 10993-5 Compliance Assessment

Report ID: #9 | Generated: 2026-08-17 05:07 UTC | Analyst: Dr. Nano Researcher

1. EXPERIMENT OVERVIEW

Experiment / Sample Name: Test ZnO Formulation Alpha
Cell Line Evaluated: A549
Medical Application: wound_healing
Lead Researcher: Dr. Nano Researcher

Nanoparticle Type: ZnO
Exposure Duration: 24h
ISO 10993-5 Compliance: PASS — Biocompatible
Data Source: Direct Entry

2. STUDY PARTICIPANT & BIOLOGICAL SAMPLE TRACEABILITY

Biological Sample: Standard in-vitro reference cell culture line. Anonymized laboratory experiment.

3. AVERAGED CYTOTOXICITY & BIOMARKER METRICS

Mean Cell Viability: 84.5%
Reactive Oxygen Species (ROS): 1.45x
Apoptosis Inducement: 2.1%

Mean ZnO Concentration: 15.0 µg/mL
LDH Membrane Leakage: 3.8%

4. MACHINE LEARNING CYTOTOXICITY ASSESSMENT

Predicted Toxicity Score: 18.2 / 100
Estimated IC50 (4PL Curve): 68.4 µg/mL
ML Model Confidence: 98.5%

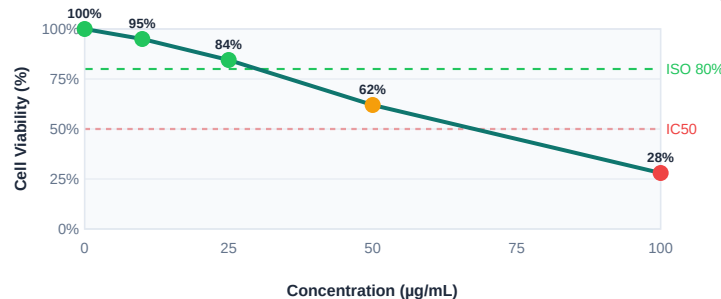
Risk Classification: Low
Biocompatible Safe Window: 0.0 – 35.0 µg/mL

5. SCIENTIFIC INTERPRETATION & RECOMMENDATION

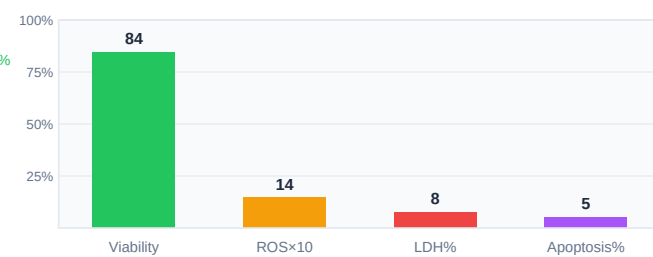
The ZnO nanoparticle formulation demonstrated acceptable biocompatibility at the evaluated concentrations. ISO 10993-5 criteria are met. Mean cell viability at 84.5% satisfies the $\geq 80\%$ threshold.

6. DOSE-RESPONSE CURVE & BIOMARKER PROFILE CHARTS

Graph A: Cell Viability vs. Concentration (Dose-Response Curve)



Graph B: Multi-Biomarker Safety Profile



7. CLINICAL SUGGESTIONS & NEXT STEPS

1. SAFE FOR USE:

Cell viability at 84.5% exceeds ISO 10993-5 threshold of $\geq 80\%$. The formulation demonstrates acceptable biocompatibility for in-vitro conditions.

2. THERAPEUTIC WINDOW:

IC50 of 68.4 µg/mL indicates a wide safe dosage range. Maintain concentrations well below IC50 — suggested clinical range: 0.0 – 35.0 µg/mL.

3. NEXT STEPS:

Proceed to Phase II in-vivo validation in an appropriate animal model (e.g., murine wound model). Evaluate systemic toxicity and immune response at escalating dose levels.

4. WOUND DRESSING SPECIFIC:

Consider hydrogel matrix incorporation at determined safe concentration. Evaluate antibacterial efficacy against *S. aureus* and *P. aeruginosa* wound pathogens.

5. REGULATORY PATH:

Compile cytotoxicity dataset per ISO 10993-5 guidance for medical device regulatory submission. Retain raw data, chain of custody records, and this report for audit compliance.